Рубежный контроль №1 по курсу ПиК ЯП

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Вариант Д8

Текст программы

```
from operator import itemgetter


class HDD:
def __init__(self, id, model, capacity, price, surname, comp_id):
self.id = id
self.model = model
self.capacity = capacity
self.price = price
self.surname = surname
self.comp_id = comp_id


class Computer:
def __init__(self, id, name):
self.id = id
self.name = name


class HDDComputer:
def __init__(self, comp_id, hdd_id):
self.comp_id = comp_id
self.hdd_id = hdd_id


computers = [
Computer(1, 'Альфа-ПК'),
Computer(2, 'Игровой ПК'),
Computer(3, 'Офисный ПК'),
Computer(4, 'Архивный ПК'),
]


hdds = [
```

```
HDD(1, 'Seagate Barracuda', 1000, 5000, 'Иванов', 1),
HDD(2, 'WD Blue', 2000, 7000, 'Петров', 1),
HDD(3, 'Toshiba P300', 1000, 4500, 'Сидорова', 2),
HDD(4, 'Samsung 870 EVO', 2000, 6000, 'Кузнецов', 2),
HDD(5, 'Kingston A400', 500, 3500, 'Иваненко', 3),
]


hdds_computers = [
HDDComputer(1, 1),
HDDComputer(1, 2),
HDDComputer(1, 3),
HDDComputer(2, 2),
HDDComputer(2, 3),
HDDComputer(3, 4),
HDDComputer(4, 1),
HDDComputer(4, 5),
HDDComputer(4, 2),
]


def main():

one_to_many = [
(h.model, h.surname, h.price, c.name)
for c in computers
for h in hdds
if h.comp_id == c.id
]

many_to_many_temp = [
(c.name, link.comp_id, link.hdd_id)
for c in computers
for link in hdds_computers
if c.id == link.comp_id
]

many_to_many = [
(h.model, h.price, comp_name)
for comp_name, comp_id, hdd_id in many_to_many_temp
for h in hdds
if h.id == hdd_id
]

print('Задание Д1')
```

```
res_11 = [
(model, comp_name)
for model, surname, price, comp_name in one_to_many
if surname.endswith('ов')
]

print(res_11)

print('\nЗадание Д2')

res_12_unsorted = []

for c in computers:

c_hdds = list(
filter(lambda i: i[3] == c.name, one_to_many)
)

if len(c_hdds) > 0:

c_prices = [price for _, _, price, _ in c_hdds]

c_prices_sum = sum(c_prices)

c_prices_avg = c_prices_sum / len(c_prices)

res_12_unsorted.append((c.name, c_prices_avg))

res_12 = sorted(
res_12_unsorted,
key=itemgetter(1),
reverse=True
)

print(res_12)

print('\nЗадание Д3')

res_13 = {}

for c in computers:

if c.name.startswith('A'):
```

```
c_hdds = list(
filter(lambda i: i[2] == c.name, many_to_many)
)

c_hdds_names = [x for x, _, _ in c_hdds]

res_13[c.name] = c_hdds_names

print(res_13)


if __name__ == '__main__':
main()
```

Результаты выполнения

Задание Д1
[('Seagate Barracuda', 'Альфа-ПК'), ('WD Blue', 'Альфа-ПК'), ('Samsung 870 EVO', 'Игровой ПК')]

Задание Д2
[('Альфа-ПК', 6000.0), ('Игровой ПК', 5250.0), ('Офисный ПК', 3500.0)]

Задание Д3
{'Альфа-ПК': ['Seagate Barracuda', 'WD Blue', 'Toshiba P300'], 'Архивный ПК': ['Seagate Barracuda', 'Kingston A400', 'WD Blue']}